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Topical Classification of Text Fragments Accounting for Their Nearest Context

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Abstract—We present an approach to topical classification of biographical text fragments that takes into account the nearest context of classified fragments using a neural network with several inputs. The choice of the model architecture is based on the assumption that since texts written in a natural language differ in consistency and coherence, the context of a passage can be used as additional input data. The model was trained and tested on the biographical corpus compiled by ourselves. The results obtained using the proposed approach outperformed the results of models that do not take into account the context of the passage.

Keywords: sentence classification, data mining, recurrent neural networks, natural language processing, biographical texts, context, text corpus, biographical research, Word2Vec, BERT

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1. INTRODUCTION

Interpreting unstructured information presented as natural language texts is one of the key challenges in data mining and information retrieval. A special case of the information retrieval task is the search for biographical information that is relevant for biographical research, collection of historical and genealogical data, and biographical facts from the life of an individual. Characteristic features of this task include, first, the genre variety of sources of biographical information (autobiography, notes, essays etc.), and, second, diversity of biographical information, including various aspects of human life: political, personal, social, and cultural.

The development of methods for searching for biographical information has been carried out mainly in two directions [1]:

1) subject, or "indirect" search, where a search engine user formulates queries based on biographical facts about a certain person known to him/her and is trying to find missing information based on them;

2) free search, where the user does not have initial information about the person of interest. Free search involves looking through biographical texts about a person in order to find specific biographical information that is relevant to the user's requirements (for example, information about professional activities or personal life).

In the latter case, the user is forced to peruse large volumes of texts. A tool for automatic processing of biographical texts that would extract from them fragments associated with various types of biographical information could help reduce the time spent on the free search for biographical information. Such a tool can be implemented based on the methods of automatic text classification. In this case, the text, previously divided into fragments, is fed to the input of the classifier, which determines the topic of each fragment.

Different texts, depending on the genre, can have a standardized structure (such texts include, for example, documents of an official business style) or have a structured structure based not on the arrangement of structural parts but on logical coherence [2]. Biographical texts can serve as an example of the second type of texts due to the fact that the information in them is usually presented in chronological order, and knowledge of the subject matter of a passage of such a text allows us to assume which fragment precedes it and which one is located after. This feature allows us to assume that taking into account the logic of presentation of biographical texts and taking into account the immediate context of the fragments will make it possible to improve the quality of the topical classification of passages.

In this work, we propose an approach to topical classification of fragments of biographical texts based on their nearest context. A sentence is considered as a fragment, since this linguistic unit is a grammatically organized combination of words (or a word) with semantic completeness [3]. In this work we compare several machine learning models for classifying biographical text fragments with and without nearest context. The experiments are carried out on a corpus of biographical texts collected by ourselves.

2. WORKS ON RELATED TOPICS

The topic of the work mainly concerns two natural language processing problems:

1) extraction of biographical information (biographical facts);

2) topical classification of sentences.

Existing works devoted to the solution of these problems pursue different practical goals and use different approaches. However, in general, in the literature on this topic information extraction and text classification are viewed as poorly formalized problems, and the methods used to solve them often depend on the specifics of the processed texts [4, 5]. Methods for searching and extracting biographical information are developing mainly in three directions: deterministic approaches based on the use of patterns and rules; approaches based on the application of machine learning methods (in particular, neural networks); hybrid approaches. Deterministic approaches show fairly high performance in many tasks, but they require the development of a large number of features that reflect the structural, semantic and lexical features of texts. Advantages of approaches based on machine learning include automatic tuning of model parameters using a variety of examples, as well as the ability not only to correlate the results of word processing with their individual characteristics, but also to reveal more complex hidden dependencies and patterns [6]. However, the implementation of approaches using machine learning methods requires one to construct training samples of texts accompanied by high-quality labeling, which is also difficult to implement in real conditions. One important trend in natural language processing is the use of pretrained models based on deep neural networks (transfer learning) [7, 8], when a pretrained model is fine-tuned to solve specific problems [9].

Deterministic approaches to the extraction of biographical information include the work [1], which describes a technology that presents a biographical fact in the form of a tree structure whose root is the type of fact (for example, "birth"), and the leaves are entities associated with the fact. The work [10] proposes an approach to retrieve biographical events based on Wikipedia traffic. In [11], a set of rules is presented for extracting biographical information from Russian texts. The work [12] compares several rule-based approaches and proposes a taxonomy of biographical facts that includes seven relationship types. There are many works whose authors have applied various machine learning methods to classify fragments of biographical texts or extract biographical facts. In particular, [13] uses a naive Bayesian classifier, [14] uses support vector machines and decision trees, and [15, 16] use neural networks. In [17], a comparison of rule-based approaches with a support vector machine was performed using the example of binary classification of phrases with and without biographical information, as a result of which the support vector machine showed significantly higher quality. The work [18] compared different types of machine learning for the extraction of relations in biographical texts (essentially fact extraction) using the example of the Portuguese language. Hybrid approaches include, e.g., [19, 20].

In many works related to the search for biographical information, experiments were carried out on Wikipedia texts (in particular, see [21–26]). This is due to the fact that Wikipedia contains a rich and varied material for research presented, however, in a standardized form.

The peculiarities of the problem of classification of sentences are, first, the relatively small length of classified texts and, second, the presence of context in sentences, which can also be taken into account by classification algorithms. Whether the context will be taken into account during classification depends on the specifics of the problem being solved and the data available for the study. Many existing systems for classification of short texts use algorithms based on the use of probabilistic and statistical methods: Bayesian classifiers [27], conditional random fields [28], hidden Markov models [29], logistic regression [30]. To solve text classification problems, recurrent neural networks trained using vector representations of symbols and words are widely used. In particular, approaches to natural language processing based on the use of recurrent neural networks have been presented in [31–35]. In recent years, tools using the ELMo and BERT models (in particular, [36–38]) have shown excellent results in the classification of short texts.

Among the research related to the use of context, we note the work [39], where a neural network architecture was proposed for classifying utterances in a dialogue. The model described in this work had several inputs, one of which received the current utterance as input, and the others received its context, i.e., preceding phrases. The model constructed in this way demonstrated higher quality of classification in comparison with a conventional recurrent neural network on English-language conversational text corpora. In [40, 41], the same team proposed a neural network model for dividing fragments of medical article annotations into five available classes: introduction, review of existing works, methodology, results and conclusions. The work [42] describes an approach to classifying sentences by sentiment using a number of discursive features.

3. METHODS

In this work, we propose a neural network architecture for the classification of fragments of biographical texts based on the architecture for the classification of utterances in a dialogue as presented in [39]. The proposed model includes several inputs that separately process the current text fragment, as well as previous and subsequent fragments. The vectors, which are the results of processing fragments in the input blocks, are combined into a common layer of the neural network. To assess classification quality, we use a corpus of biographical texts that was collected and labeled by the author in semi-automatic mode [43]. We consider two variants of the neural network's architecture, using different types of sentence representation:

1) recurrent neural network; here the text is presented as a sequence of words, and pretrained vectors of the Word2Vec model [44] are used as a matrix of vector representations of words (for the Embedding layer);

2) a feedforward network trained on vector representations of sentences obtained using the BERT model [7].

3.1. Architecture

The recurrent model is based on the use of recurrent long short-term memory (LSTM) layers, which, unlike classical recurrent architectures, provide a mechanism for storing long-term dependencies, which avoids the problem of vanishing gradients [45]. The structure of the LSTM network cell is shown in Fig. 1 [46].

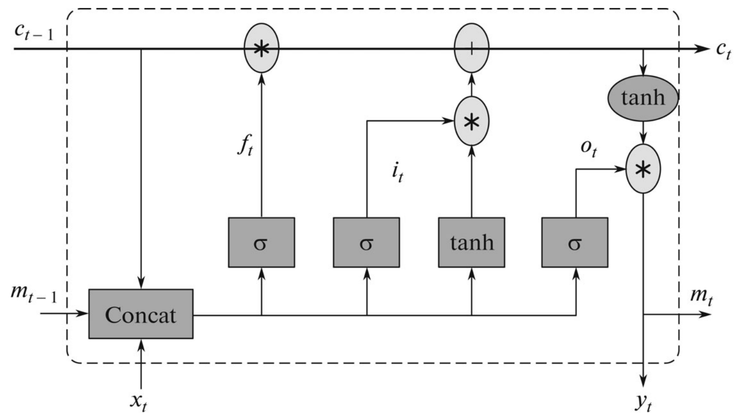


Fig. 1. LSTM cell structure.

Let x_t and y_t denote respectively the input and output signals at time t, and let c_t and m_t be the state of the cell and output at time t. Transformation of the input signal to the output signal occurs as follows:

$$\begin{gathered}
i_t = \sigma(W_{ix}x_t + W_{im}m_{t-1} + W_{ic}c_{i-1} + b_i),\\
f_t = \sigma(W_{fx}x_t + W_{fm}m_{t-1} + W_{fc}c_{i-1} + b_f),\\
o_t = \sigma(W_{ox}x_t + W_{om}m_{t-1} + W_{oc}c_{i-1} + b_o),\\
m_t = o_t \odot h(c_t),\\
y_t = \varphi(W_{ym}m_t + b_y),\\
c_t = f_t \odot c_{t-1} + i_t \odot g(W_{cx}x_t + W_{cm}m_{t-1} + b_c),
\end{gathered} \tag{1}$$

where W_{cx}, W_{ix}, W_{fx}, W_{ox} are input weights, W_{cm}, W_{im}, W_{fm}, W_{om} are cell state weights, b_o, b_i, b_f are bias vectors, W_{ic}, W_{fc}, W_{oc} are link weights between cells and output filter layer, W_{ym} and b_y are the weight and bias for the output, and σ, g, h are some nonlinear functions.

In a feedforward network, recurrent layers are replaced by feedforward layers, i.e., layers without recurrent connections, with the hyperbolic tangent activation function.

The input data for the models includes the current sentence and its context, i.e., n preceding and n subsequent sentences. Let s_j denote the sentence number j. Then the input is a set of sentences S:

$$S = \{s_{j-n}, \ldots, s_{j-1}, s_j, s_{j+1}, \ldots, s_{j+n}\}, \quad j \in [n+1, J-n], \tag{2}$$

where J is the number of sentences in the text.

In the event that $j < n+1$ or $j > J-n$, the context suggestion s_k, $k \in \{j-n, \ldots, j-1, j+1, \ldots, j+n\}$ is fed to the network input if $1 \leqslant k \leqslant J$. Otherwise, the markers for the beginning (for $k < 1$) or end (for $k > J$) of the text are used as input for the corresponding position.

Each sentence from the set S corresponds to a separate network input. Thus, the input data of the network consists of $2n+1$ sentences, and the output of the input blocks are vectors corresponding to input sentences.

3.2. Taking Context into Account

In what follows we consider three different options for the architecture of each model.

1. In the first case, concatenated output vectors of the input blocks are fed to the forward propagation layer. The resulting values go to the output layer of the model, which is also a feedforward layer with dimension equal to the number of classes and a softmax activation function. The output layer of the network returns the probability distribution among topical classes for the sentence (Fig. 2a).

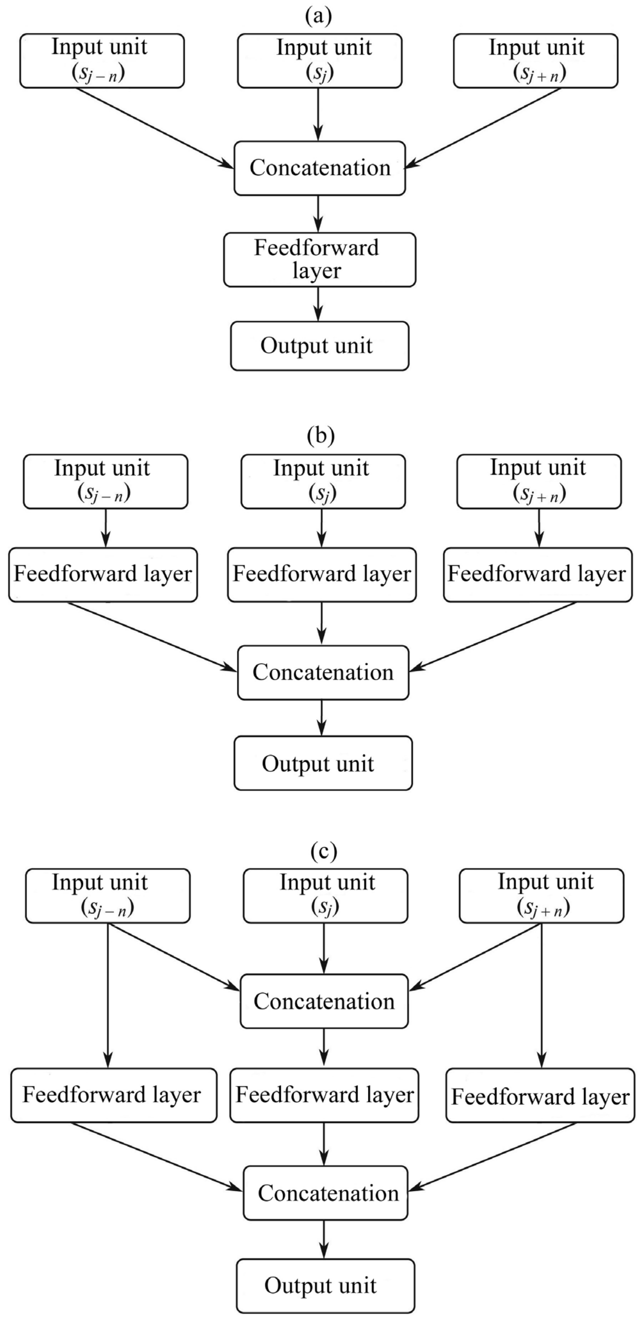


Fig. 2. Variations of the model architecture.

2. In the second case, one forward propagation layer is added to each input block, after which concatenation is carried out, and its result is fed to the output layer. Thus, the influence of the context is taken into account at the level of the last layer of the model (Fig. 2b).

3. The third case is a combination of the first two. Output vectors of the input blocks are concatenated and the result is processed by a forward propagation layer. At the same time, the output vectors of the input blocks corresponding to the context sentences are fed to the input of the feedforward layers. All resulting vectors are concatenated, and the result is fed to the output layer. Thus, the influence of the context is taken into account both at the level of the output vectors of the recurrent blocks and at the level of the last layer of the model (Fig. 2c).

4. EXPERIMENTS

4.1. Dataset

A corpus of biographical texts was compiled for training and testing the models. It is a collection containing biographical texts from the online encyclopedia *Wikipedia*, broken down into sentences and provided with topical labels. The version of the corpus used for the experiments contains 200 texts describing the biographies of people who lived or are living in the XX and XXI centuries. The corpus is freely available on the website [47].

Each sentence in the corpus of biographical texts is associated with a class label that most fully corresponds to its subject matter: birth, information about the parental family, place of residence, occupation, place of work, family, education, personal events, professional events, death. Some sentences in the corpus are labeled with two classes, main and auxiliary. In this work, when classifying such sentences, the main class label was used. Thus, each fragment corresponds to one of 10 classes. Characteristics of the corpus are presented in Table 1.

To equalize the number of examples in classes, simple oversampling was carried out, i.e., random duplication of elements in minority classes. The total number of elements for training and validation of the models after oversampling is 8251, and the size of the test sample where the final quality of the models was assessed was 177 sentences. Data preprocessing included converting text to lower case, removing special characters, stop words, and punctuation marks, and converting words to their original form.

Table 1. Corpus characteristics

Class	Average sentence length (tokens)	Average context length for $n = 1$ (tokens)	Number of samples
Birth	13.9	13.25	134
Parental family	13.05	28.9	86
Place of residence	13.17	31.23	94
Occupation	16.93	32.4	943
Work place	14.4	32.27	113
Family	11.83	24.25	48
Education	15.73	30.04	374
Persional events	20.35	38.07	105
Professional events	21.36	37.72	490
Death	10.56	22.15	111
All sentences	**16.83**	**31.52**	**2498**

In the absence of the required number of neighboring sentences in the text, special markers "begin" and "end" were supplied to the input of the models for the preceding $(s_{j-1}, s_{j-2}, \ldots, s_{j-n})$ and subsequent $(s_{j+1}, s_{j+2}, \ldots, s_{j+n})$ sentences respectively. For example, in case $j = 1$ (the sequence number of the sentence in the text), $J = 3$ (the number of sentences in the text) and $n = 3$ (the size of the context), the input data will look as follows:

$$\begin{aligned}
s_{j-3} &= \text{“\textit{begin},”} \\
s_{j-2} &= \text{“\textit{begin},”} \\
s_{j-1} &= \text{“\textit{begin},”} \\
s_j &= \text{“Text of sentence } s_j\text{,”} \\
s_{j+1} &= \text{“Text of sentence } s_{j+1}\text{,”} \\
s_{j+2} &= \text{“Text of sentence } s_{j+2}\text{,”} \\
s_{j+3} &= \text{“\textit{end}.”}
\end{aligned}$$

4.2. Implementation and Training of Models

During the experiments, we have compared context-aware models with neural networks based on the Transformer architecture, as well as with support vector machines tested on the representations of sentences in the form of Bag-of-Words with TF-IDF weights. In this work, we used two BERT models:

1) mBERT (multilingual BERT) supporting 104 languages [7];
2) RuBERT, a BERT model for the Russian language trained on the Russian-language Wikipedia and news portal texts [48]. For Russian-language texts, this model has shown quality that significantly exceeds the quality of the multilingual BERT model on a number of tasks.

Models based on the Transformer architecture are implemented using the Transformers [49] and PyTorch [50] libraries and the Python 3.6 programming language. The input data for the BERT model is a sentence enclosed in [CLS] and [SEP] tokens. The sentence is processed by a tokenizer that converts tokens into a sequence of indices according to the model vocabulary. The input vector size for BERT is limited to 512 tokens, the batch size is 8, and the number of training epochs is 3.

The support vector machine is implemented using the Scikit-Learn (LinearSVC) [51] library. As input data, we used representations of sentences according to the Bag-of-Words model with TF-IDF weights (the bag-of-words matrix, where the value of the TF-IDF measure for a given word in a given document is located at the intersection of the corresponding row and column). The dimension of vector representations is 5000 features.

The implementation of context-aware models was done using the Keras [52] library and the Python 3.6 programming language. Input blocks of recurrent networks consist of an input layer, an embedding layer, and a recurrent LSTM layer. The dimension of the LSTM layer is 64 neurons, forward propagation layers have 32 neurons. Input blocks of feedforward networks include an input layer and one feedforward layer. The dimension of the forward propagation layers in the input blocks is 256 neurons, in the common layer, 32 neurons. The activation function for the inner layers is hyperbolic tangent, for the output layer, softmax. Optimization of model hyperparameters was carried out on the example of context-free models using a simple grid search. The list of parameters to be optimized and their ranges is given in Table 2. Adaptive moment estimation (Adam optimizer) is used as an optimization algorithm for all models, and categorical cross-entropy is used as the loss function.

Figure 3 shows the schemes of context-free recurrent model and feedforward network. Input blocks are marked in italics. In context-aware models (with $n > 0$), each input sentence from the set S corresponds to a separate input block. Figure 4 shows a visualization of three variations of the recurrent model for $n = 1$ as an example. Models for $n > 1$ have a similar form with more inputs.

Table 2. Optimizing model parameters

Parameter	Set of parameters	Chosen value
Activation function on internal layers (recurrent network)	hyperbolic tangent, relu	hyperbolic tangent
Activation function on internal layers (feedforward network)	hyperbolic tangent, relu	hyperbolic tangent
Dimension of the LSTM layer (recurrent network)	of degree 2 from the range [32;256]	64
Dimension of the feedforward layer in the input unit (feedforward network)	of degree 2 from the range [64;512]	256
Dimension of the common feedforward layer (recurrent network)	of degree 2 from the range [32;128]	32
Dimension of the common feedforward layer (feedforward network)	of degree 2 from the range [32;128]	32

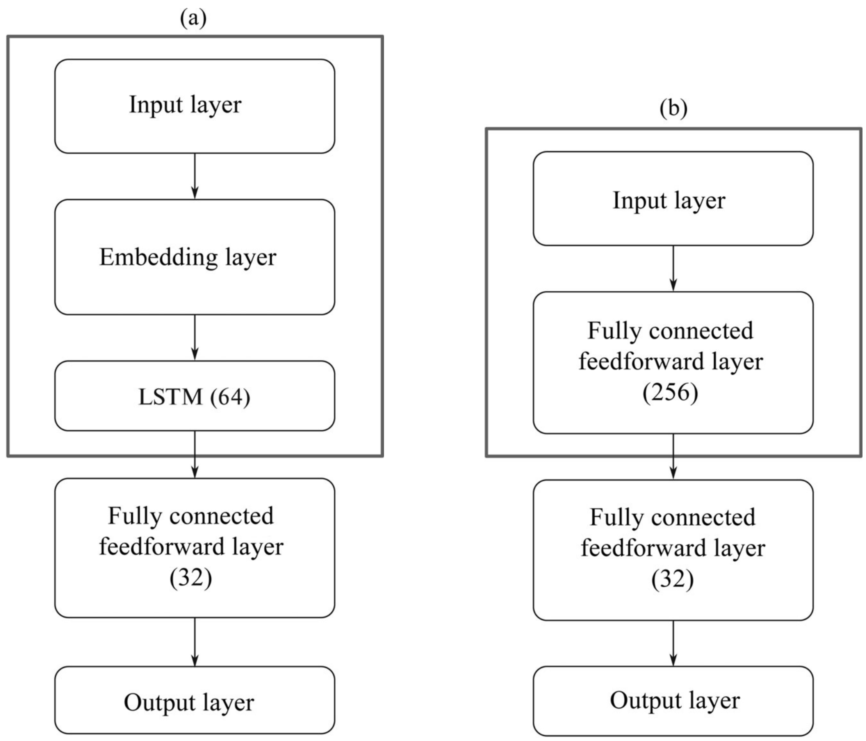


Fig. 3. Context-free models: (a) recurrent network; (b) feedforward network.

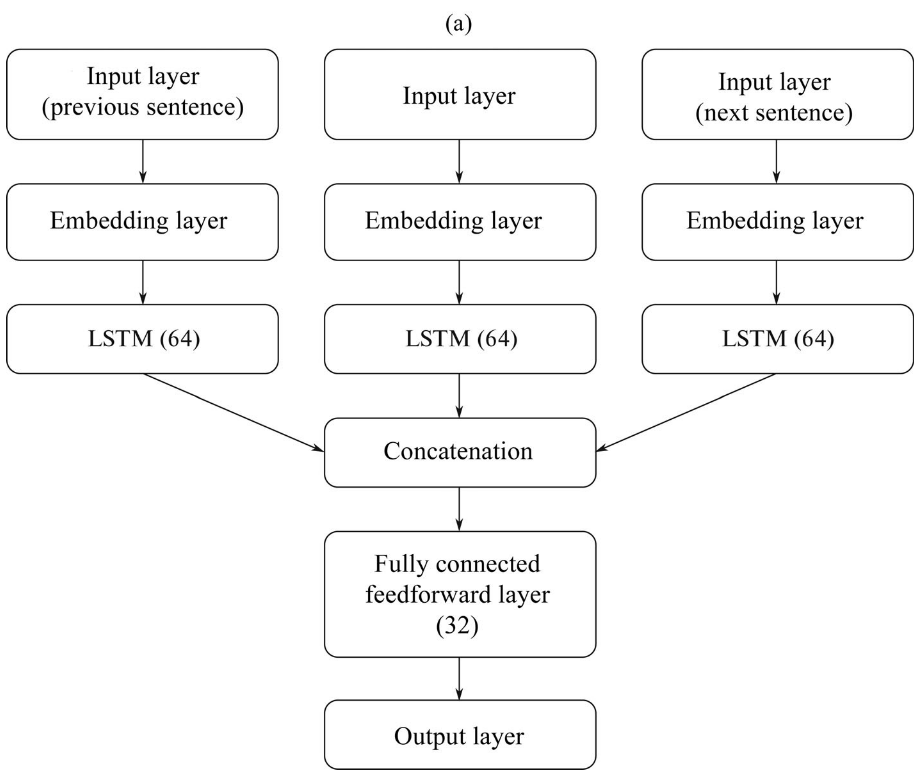
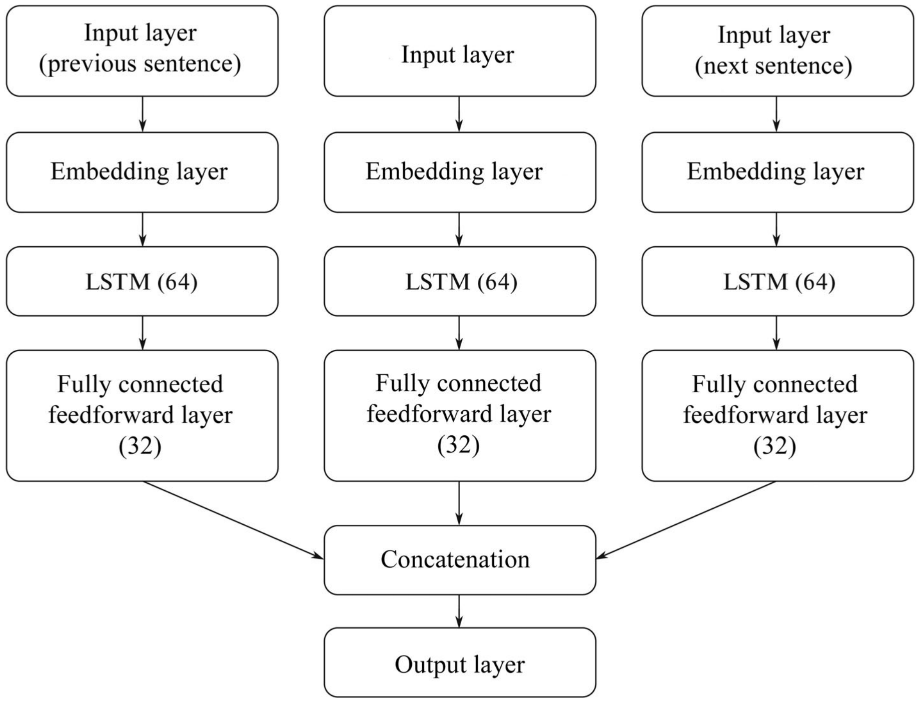


Fig. 4. Visualization of recurrent models for $n = 1$.

(b)



(c)

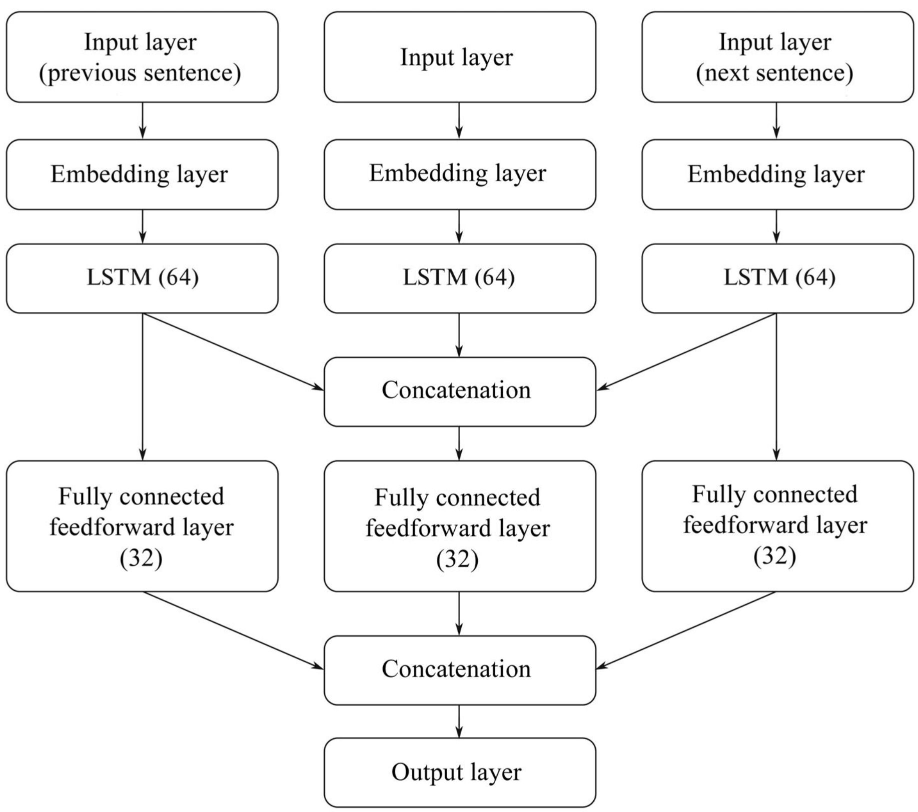


Fig. 4. (Contd.)

The dataset for training neural network models was divided into training and validation samples. Training was carried out using the training set, and the training was stopped according to the model indicators on the validation set. The final testing of the model was carried out on an independent test sample that did not participate in the training process. For recurrent networks, the input data were fed into the model in the form of sequences of words based on a matrix of vector word representations composed of vectors of the Word2Vec model and a set of tokens presented in the training set. As a Word2Vec model, we used a model trained on the texts of the Russian-language Wikipedia and the National Corpus of the Russian language for 2018 using the Skip-gram learning algorithm [53]. The dimension of the vector representation of the word in this model is 300. The input data of feedforward networks looks like a one-dimensional vector of 768 dimensions, obtained for the current text fragment from the RuBERT model using the DeepPavlov library [54].

The source code for all models is available at [55].

4.3. Results

To assess the results, we computed the F-measure (macro-averaging), defined as the average value of the F-measure values calculated for each class in terms of precision and recall. The values of precision and completeness are shown in brackets after the F-measure value (the first number shows precision, the second, recall).

Table 3 shows the classification quality metrics. Since, due to the random initialization of the initial parameters, classification results may vary with different model runs, each neural network model was run m times, and the table shows the average values. In this work $m = 5$. In our ex-

Table 3. Model evaluation (F-measure (Precision / Recall), values shown in %)

Architecture	Value of n			
	0	1	2	3
Recurrent (variant 1)	–	91.25 (90.31 / 92.15)	93.13 (91.23 / 94.13)	**94.77 (91.33 / 95.77)**
Recurrent (variant 2)	–	92.54 (91.45 / 92.56)	93 (91.98 / 94.03)	92.9 (91.56 / 92.35)
Recurrent (variant 3)	–	92.07 (92.01 / 91.87)	92.3 (92.12 / 91.96)	94.07 (91.4 / 95.88)
Feedforward (variant 1)	–	86.23 (85.89 / 88.42)	87.14 (84.78 / 89.2)	87.45 (85.91 / 88.45)
Feedforward (variant 2)	–	87.18 (86.56 / 89.12)	87.03 (86.22 / 88.14)	87.5 (87.02 / 88.49)
Feedforward (variant 3)	–	87.35 (87.53 / 89.36)	87.56 (87.34 / 89.01)	87.14 (87.02 / 88.32)
Recurrent (context-free)	89.46 (90.13 / 88.3)	–	–	–
Feedforward (context-free)	86.83 (89.3 / 88.26)	–	–	–
LinearSVC	66.37 (64.39 / 77.44)			
mBERT	89.01 (92.11 / 88.12)	–	–	–
RuBERT	**93.16 (90.72 / 97.05)**	–	–	–

Table 4. Sample model errors

Fragment	Labeling	Classification result (RuBERT)	Classification result (recurrent context-aware model)
s_{j-3}: "begin" s_{j-2}: "begin" s_{j-1}: "begin" s_j: "Born in the family of an agronomist from a family of Russian Germans." s_{j+1}: "Two of his younger sisters, Ksenia (1894–1955) and Maria (1897–1942), also became painters." s_{j+2}: "In 1905–1907, he took private painting lessons from I.Ya. Bilibin." s_{j+3}: "In 1911, he became close with M.V. Matyushin and E.G. Guro, often visiting at their flat in a house on Pesochnaya street."	Parental family information	Birth	Birth
s_{j-3}: "begin" s_{j-2}: "begin" s_{j-1}: "Studied at school no. 1, industrial college." s_j: "1935—aviaclub, then civil aviation in Georgia with her husband." s_{j+1}: "1941—instructor (200 cadets)." s_{j+2}: "April 1944—Saransk, 3rd Naval Flight School (attack pilots)." s_{j+3}: "Afterwards—assigned to 7 GvPShAP Air Force KBF (regiment commander, twice Hero of the Soviet Union, A.E. Mazurenko)."	Occupation	Family	Occupation
s_{j-3}: "In 1918, after demobilisation, he entered the Petrograd State Free Art Workshops, studied with K.S. Petrov-Vodkin, then with Matyushin." s_{j-2}: "After completing his studies in 1923, he continued to work under Matyushin in the Department of Organic Culture of Inkhuk, and joined the Zorved group he created." s_{j-1}: "In the 1920s, he took an active part in exhibitions of the 'workshop of spatial realism.'" s_j: "Got acquainted with K.S. Malevich, N.M. Suetin, I.N. Khardzhiev, I.G. Ehrenburg, maintained constant correspondence with them." s_{j+1}: "In 1927 he moved to Moscow." s_{j+2}: "In the 1930s worked a lot on monumental art." s_{j+3}: "Took part in the design of the USSR pavilion at the International Exhibition in Paris (1937)."	Personal events	Personal events	Professional events

periments, we considered context-aware models in the range $0 \leqslant n \leqslant 3$, since a further increase in the value of n did not give an increase in the quality of classification and negatively affected the time complexity of the model.

The best values of F-measure among all considered models (recurrent in variant 1, 94.77%) and among context-free models (RuBERT, 93.16%) are highlighted in bold in the table. As the data in the table shows, in most cases addition of context sentences has improved the quality of sentence classification. Moreover, for recurrent models the best result was achieved with $n = 3$, and for feedforward networks with $n = 2$. We see the largest absolute rates of improvement for recurrent models (+5.31%).

Table 4 shows examples of sentences erroneously classified by the recurrent context-aware model and the RuBERT model. In most cases, mistakes are related to fragments that are topically related to more than one class. Many of these fragments were labeled with an auxiliary class in the original corpus. In particular, the sentence from the first example (a fragment of the biography of the artist B.V. Ender) was assigned the main class "Information about the parental family" and auxiliary class "Birth" by the corpus labelers. The choice of the main class label is due to the fact that the sentence describes the person's origin and does not specify the fact of birth (date and place). Both networks categorized this sentence into the "Birth" class. The second sentence is an example of the strength of context-aware models. Probably, the fragment describing the professional activities of the Soviet pilot L.I. Shulaykina and mentioning her spouse is correctly assigned to the class "Occupation" due to the subject of the context, while the RuBERT model classified this fragment as an element of the class "Family." An opposite example is the third sentence (a fragment of the biography of B.V. Ender), which was classified by the labelers as "Personal events," since it describes the artist's meetings and personal acquaintances. The model, taking into account the context, classified this fragment as "Professional events" (in the corpus, this class corresponds to references to official meetings and awards). Perhaps the result is due to the "professional" theme of the context.

5. CONCLUSION

In this work, we have presented an approach to the topical classification of text fragments that takes into account their immediate context. The model was tested with a dataset of the corpus of biographical texts. Since biographical texts differ in chronological order of presentation, all models that take the context of a passage as input performed better than context-free models.

The architecture proposed in this work can be used to solve similar problems of topical classification of text passages that have an explicit logical structure and sequence of presentation.

FUNDING

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